

CHRISTOPHER OOI EN SHEN

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Education

Wilfrid Laurier University

Sep. 2023 – Jan. 2027

Bachelor of Science in Computer Science

Waterloo, Ontario

- CGPA of **3.85**
- **Relevant Coursework:** Operating Systems, Data Structures & Algorithms, Object Oriented Programming, System Programming, Computer Networks, Database Systems

Work Experience

Manulife 🌐

May. 2026 - Sep. 2026

Incoming Software Engineer Intern

Waterloo, Ontario

Hecaton 🌐

Oct. 2025 – Present

Contract Software Engineer

Waterloo, Ontario

- Delivered a production-ready **Bitcoin notarization platform** by engineering a secure **full-stack** architecture (**NextJs**, **Supabase**, **Prisma**, **bitcoinjs-lib**), successfully scaling to process **500+** monthly transactions post-launch.
- Delivered a **production frontend foundation** for **Fleetcraft** (Series A) by owning the **end-to-end UI architecture** and implementing a **hybrid search capability** using **BM25** + vector embeddings.
- Shipping **reAIgents** (AI real estate platform) handling **3.1M+** property records, saving **\$10–15K** in API costs via **protected geocoding** with caching and rate-limiting; architected **Next.js 15 service layer**, implemented **PostGIS spatial queries**, and built **real-time notification system**.

Bolttech 🌐

May. 2025 – Sep. 2025

Data Engineer Intern

Kuala Lumpur, Malaysia

- Generated **\$15000+** in **annual savings** by **re-architecting legacy automation** into a **serverless AWS ETL pipeline (Lambda/Step Functions)** scaling to process **1M+** daily inventory records.
- Reduced data processing time by **95% (45m → 2m)** by engineering memory-efficient **Pandas + Databricks** pipelines for large datasets and optimizing transformation patterns to avoid memory bottlenecks.

Sensoft Technologies 🌐

May. 2024 – Sep. 2024

Software Engineer Intern

Penang, Malaysia

- Delivered a real-time React monitoring dashboard for **15,000+ Modbus sensors** while maintaining **60 FPS rendering** by implementing list virtualization and optimizing high-frequency WebSocket UI updates.
- Reduced energy costs by **15%** by building a hybrid forecasting + reporting pipeline using **Facebook Prophet** for time-series prediction and the **OpenAI API** to generate automated, actionable reports.
- Achieved **100ms latency** for real-time people detection on edge devices by parallelizing frame capture and **YOLOv8** inference through a multi-threaded video processing pipeline.

Personal Projects

MentorTrader 🌐 | Next.js, PostgreSQL, Stripe, AWS, Sentry

- Created a **marketplace platform** focused on connecting **day traders with credible online trading mentors**.
- Onboarded **120+ active mentors** and **2,000+ monthly visitors**, featuring management **dashboards** and **partnerships** with companies like **Darwinex**, **Whop**.
- **Deployed** a **Next.js** application on **Vercel**, using **Supabase** for **secure authentication** and **database**, **Redis** for caching, **Sentry** for observability, and **Stripe** for payments.

Faves 🌐 | Express.js, Firebase, TailwindCSS, Google Cloud, OpenAI

- Built a discovery **platform** aggregating **1,000+ AI-verified recommendations** from influential figures.
- Features an **automated content pipeline** where public submissions trigger **LLM-based citation generation** followed by a rigorous **admin moderation** workflow.
- Architected a **serverless backend** using **Google Cloud Functions** and **Node.js**, securing public endpoints via **Joi** validation, **rate-limiting**, and **reCAPTCHA** to prevent abuse.

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, Java, C, Ruby

Frameworks: Next.js, React, Express, FastAPI, Flask, Rails

Data/Infra: AWS (Lambda, Step Functions), Databricks, Docker, Redis, PostgreSQL

Developer Tools: GitHub Actions, Sentry, Vercel, GCP Functions